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# Vivado v2025.2 (64-bit)
# SW Build 6299465 on Fri Nov 14 19:35:11 GMT 2025
# IP Build 6300035 on Fri Nov 14 10:48:45 MST 2025
# SharedData Build 6298862 on Thu Nov 13 04:50:51 MST 2025
# Start of session at: Wed Jul 22 01:47:30 2026
# Process ID           : 37816
# Current directory    : D:/Vivado/UART/UART.runs/synth_1
# Command line         : vivado.exe -log uart_top.vds -product Vivado
-mode batch -messageDb vivado.pb -notrace -source uart_top.tcl
# Log file             :
D:/Vivado/UART/UART.runs/synth_1/uart_top.vds
# Journal file         : D:/Vivado/UART/UART.runs/synth_1\vivado.jou
# Running On           : AK_LAPTOP
# Platform             : Windows Server 2016 or Windows 10
# Operating System     : 26200
# Processor Detail     : Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz
# CPU Frequency        : 2496 MHz
# CPU Physical cores   : 4
# CPU Logical cores    : 8
# Host memory          : 8419 MB
# Swap memory          : 14495 MB
# Total Virtual        : 22914 MB
# Available Virtual    : 2323 MB
#-----
source uart_top.tcl -notrace
create_project: Time (s): cpu = 00:00:10 ; elapsed = 00:00:09 .
Memory (MB): peak = 525.285 ; gain = 219.844
Command: read_checkpoint -auto_incremental -incremental
D:/Vivado/UART/UART.srscs/utils_1/imports/synth_1/uart_top.dcp
INFO: [Vivado 12-5825] Read reference checkpoint from
D:/Vivado/UART/UART.srscs/utils_1/imports/synth_1/uart_top.dcp for
incremental synthesis
INFO: [Vivado 12-7989] Please ensure there are no constraint
changes
Command: synth_design -top uart_top -part xc7a15tcp236-1
Starting synth_design
```

Attempting to get a license for feature 'Synthesis' and/or device
'xc7a15t'

INFO: [Common 17-349] Got license for feature 'Synthesis' and/or
device 'xc7a15t'

INFO: [Device 21-403] Loading part xc7a15tcp236-1

INFO: [Designutils 20-5440] No compile time benefit to using
incremental synthesis; A full resynthesis will be run

INFO: [Designutils 20-4379] Flow is switching to default flow due
to incremental criteria not met. If you would like to alter this
behaviour and have the flow terminate instead, please set the
following parameter config_implementation

{autoIncr.Synth.RejectBehavior Terminate}

INFO: [Synth 8-7079] Multithreading enabled for synth_design using
a maximum of 2 processes.

INFO: [Synth 8-7078] Launching helper process for spawning children
vivado processes

INFO: [Synth 8-7075] Helper process launched with PID 37936

Starting RTL Elaboration : Time (s): cpu = 00:00:07 ; elapsed =
00:00:09 . Memory (MB): peak = 1259.793 ; gain = 543.539

INFO: [Synth 8-6157] synthesizing module 'uart_top'

[D:/Projects/UART/codes/uart_top.v:3]

INFO: [Synth 8-6157] synthesizing module 'brg'

[D:/Projects/UART/codes/brg.v:3]

Parameter CLOCK_FREQ bound to: 100000000 - type: integer

Parameter BAUD_RATE bound to: 9600 - type: integer

INFO: [Synth 8-6155] done synthesizing module 'brg' (0#1)

[D:/Projects/UART/codes/brg.v:3]

INFO: [Synth 8-6157] synthesizing module 'tx'

[D:/Projects/UART/codes/tx.v:3]

INFO: [Synth 8-6155] done synthesizing module 'tx' (0#1)

[D:/Projects/UART/codes/tx.v:3]

INFO: [Synth 8-6157] synthesizing module 'rx'

[D:/Projects/UART/codes/rx.v:3]

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INFO: [Synth 8-6155] done synthesizing module 'rx' (0#1)
[D:/Projects/UART/codes/rx.v:3]
INFO: [Synth 8-6155] done synthesizing module 'uart_top' (0#1)
[D:/Projects/UART/codes/uart_top.v:3]
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Finished RTL Elaboration : Time (s): cpu = 00:00:10 ; elapsed =
00:00:12 . Memory (MB): peak = 1379.121 ; gain = 662.867
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Start Handling Custom Attributes
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Finished Handling Custom Attributes : Time (s): cpu = 00:00:10 ;
elapsed = 00:00:12 . Memory (MB): peak = 1379.121 ; gain = 662.867
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Finished RTL Optimization Phase 1 : Time (s): cpu = 00:00:10 ;
elapsed = 00:00:12 . Memory (MB): peak = 1379.121 ; gain = 662.867
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Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.003 . Memory (MB): peak = 1379.121 ; gain = 0.000
INFO: [Project 1-570] Preparing netlist for logic optimization

Processing XDC Constraints
Initializing timing engine
Parsing XDC File
[D:/Vivado/UART/UART.srscs/constrs_1/new/uart_constraints.xdc]
Finished Parsing XDC File
[D:/Vivado/UART/UART.srscs/constrs_1/new/uart_constraints.xdc]
```

INFO: [Project 1-236] Implementation specific constraints were found while reading constraint file [D:/Vivado/UART/UART.srscs/constrs_1/new/uart_constraints.xdc]. These constraints will be ignored for synthesis but will be used in implementation. Impacted constraints are listed in the file [.Xil/uart_top_propImpl.xdc].

Resolution: To avoid this warning, move constraints listed in [.Xil/uart_top_propImpl.xdc] to another XDC file and exclude this new file from synthesis with the used_in_synthesis property (File Properties dialog in GUI) and re-run elaboration/synthesis.

Completed Processing XDC Constraints

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00 . Memory (MB): peak = 1463.633 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed.

Constraint Validation Runtime : Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.009 . Memory (MB): peak = 1463.633 ; gain = 0.000

INFO: [Designutils 20-5440] No compile time benefit to using incremental synthesis; A full resynthesis will be run

INFO: [Designutils 20-4379] Flow is switching to default flow due to incremental criteria not met. If you would like to alter this behaviour and have the flow terminate instead, please set the following parameter config_implementation {autoIncr.Synth.RejectBehavior Terminate}

Finished Constraint Validation : Time (s): cpu = 00:00:19 ; elapsed = 00:00:22 . Memory (MB): peak = 1463.633 ; gain = 747.379

Start Loading Part and Timing Information

Loading part: xc7a15tcpg236-1

Finished Loading Part and Timing Information : Time (s): cpu =
00:00:19 ; elapsed = 00:00:22 . Memory (MB): peak = 1463.633 ; gain
= 747.379

Start Applying 'set_property' XDC Constraints

Finished applying 'set_property' XDC Constraints : Time (s): cpu =
00:00:19 ; elapsed = 00:00:22 . Memory (MB): peak = 1463.633 ; gain
= 747.379

INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in
module 'tx'

INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in
module 'rx'

State | New Encoding |
Previous Encoding

IDLE	000
000	
START	001
001	
DATA	010
010	
PARITY	011

```

                                011
                                STOP |                                100 |
                                100
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INFO: [Synth 8-3354] encoded FSM with state register 'state_reg'
using encoding 'sequential' in module 'tx'
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                                State |                                New Encoding |
                                Previous Encoding
-----
-----
                                IDLE |                                000 |
                                000
                                START |                                001 |
                                001
                                DATA |                                010 |
                                010
                                PARITY |                                011 |
                                011
                                STOP |                                100 |
                                100
-----
-----
INFO: [Synth 8-3354] encoded FSM with state register 'state_reg'
using encoding 'sequential' in module 'rx'
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Finished RTL Optimization Phase 2 : Time (s): cpu = 00:00:19 ;
elapsed = 00:00:23 . Memory (MB): peak = 1463.633 ; gain = 747.379
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Start RTL Component Statistics
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Detailed RTL Component Info :

+---Adders :

2 Input 3 Bit Adders := 2

+---XORs :

8 Bit Wide XORs := 2

+---Registers :

8 Bit Registers := 3

3 Bit Registers := 2

1 Bit Registers := 10

+---Muxes :

5 Input 8 Bit Muxes := 1

2 Input 3 Bit Muxes := 2

5 Input 3 Bit Muxes := 4

2 Input 1 Bit Muxes := 5

5 Input 1 Bit Muxes := 20

Finished RTL Component Statistics

Start Part Resource Summary

Part Resources:

DSPs: 45 (col length:60)

BRAMs: 50 (col length: RAMB18 60 RAMB36 30)

Finished Part Resource Summary

Start Cross Boundary and Area Optimization

Finished Cross Boundary and Area Optimization : Time (s): cpu =
00:00:22 ; elapsed = 00:00:25 . Memory (MB): peak = 1463.633 ; gain
= 747.379

Start Applying XDC Timing Constraints

Finished Applying XDC Timing Constraints : Time (s): cpu = 00:00:27
; elapsed = 00:00:31 . Memory (MB): peak = 1568.020 ; gain =
851.766

Start Timing Optimization

Finished Timing Optimization : Time (s): cpu = 00:00:28 ; elapsed =
00:00:32 . Memory (MB): peak = 1599.059 ; gain = 882.805

Start Technology Mapping

Finished Technology Mapping : Time (s): cpu = 00:00:28 ; elapsed =
00:00:32 . Memory (MB): peak = 1599.059 ; gain = 882.805

Start IO Insertion

Start Flattening Before IO Insertion

Finished Flattening Before IO Insertion

Start Final Netlist Cleanup

Finished Final Netlist Cleanup

Finished IO Insertion : Time (s): cpu = 00:00:33 ; elapsed =
00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Start Renaming Generated Instances

Finished Renaming Generated Instances : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Start Rebuilding User Hierarchy

Finished Rebuilding User Hierarchy : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Start Renaming Generated Ports

Finished Renaming Generated Ports : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Start Handling Custom Attributes

Finished Handling Custom Attributes : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Start Renaming Generated Nets

Finished Renaming Generated Nets : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Start Writing Synthesis Report

Report BlackBoxes:

+-+-----+-----+		
	BlackBox name	Instances
+-+-----+-----+		
+-+-----+-----+		

Report Cell Usage:

+-----+-----+		
	Cell	Count
+-----+-----+		
1	BUFG	1
2	CARRY4	4
3	LUT1	1

4	LUT2		7
5	LUT3		5
6	LUT4		19
7	LUT5		9
8	LUT6		19
9	MUXF7		1
10	FDRE		59
11	FDSE		1
12	IBUF		11
13	OBUF		14

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Finished Writing Synthesis Report : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

Synthesis finished with 0 errors, 0 critical warnings and 0
warnings.

Synthesis Optimization Runtime : Time (s): cpu = 00:00:21 ; elapsed
= 00:00:35 . Memory (MB): peak = 1828.230 ; gain = 1027.465

Synthesis Optimization Complete : Time (s): cpu = 00:00:33 ;
elapsed = 00:00:37 . Memory (MB): peak = 1828.230 ; gain = 1111.977

INFO: [Project 1-571] Translating synthesized netlist

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.001 . Memory (MB): peak = 1837.441 ; gain = 0.000

INFO: [Netlist 29-17] Analyzing 5 Unisim elements for replacement

INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU
seconds

INFO: [Project 1-570] Preparing netlist for logic optimization

INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00 . Memory (MB): peak = 1841.078 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Synth Design complete | Checksum: 8d64c4c

INFO: [Common 17-83] Releasing license: Synthesis
33 Infos, 0 Warnings, 0 Critical Warnings and 0 Errors encountered.
synth_design completed successfully
synth_design: Time (s): cpu = 00:00:37 ; elapsed = 00:00:55 .
Memory (MB): peak = 1841.078 ; gain = 1309.969
Write ShapeDB Complete: Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.007 . Memory (MB): peak = 1841.078 ; gain = 0.000
INFO: [Common 17-1381] The checkpoint
'D:/Vivado/UART/UART.runs/synth_1/uart_top.dcp' has been generated.
INFO: [Vivado 12-24828] Executing command : report_utilization
-file uart_top_utilization_synth.rpt -pb
uart_top_utilization_synth.pb
INFO: [Common 17-206] Exiting Vivado at Wed Jul 22 01:48:40 2026...